Kennesaw State University 2026-2027 Undergraduate Catalog

Courses

Contract All Courses |

Computer Science Engineering

- ### CSE 1101: Emerging Technology and Artificial Intelligence

2 Credit Hours

Description
This course provides a survey of computing. Particular topics include Problem Solving, Computational Thinking, Hardware, Operating Systems, Cloud Computing, Cybersecurity, Artificial Intelligence, Data Analysis, Web Development, and provides an introduction to programming in Python. A companion hands-on lab, CSE 1101L, is available and may be taken independently for students seeking applied experience.

Course Learning Outcomes
Students who successfully complete this course will be able to:

1. Demonstrate an understanding of how computers work, including an overview of hardware, logic gates, binary, and operating systems.
2. Demonstrate an understanding of cloud computing, cyber security principles, basic networking.
3. Demonstrate an understanding of Data Analysis using spreadsheets, databases and data visualization.
4. Demonstrate an understanding of the basics of programming.
5. Use computational thinking to ask questions and find different ways to solve problems.

- ### CSE 1101L: Artificial Intelligence and Technology for Innovation Lab

1 Credit Hours

Description
This lab course will allow students to explore various computing fields in real world scenarios. Topics may include Problem-solving, Computational Thinking, Hardware, Operating Systems, Cloud Computing, Cybersecurity, Artificial Intelligence, Data Analysis, Web Development, and provides an introduction to

programming in Python. This hands-on lab complements CSE 1101 lecture but may be taken independently; students who wish to gain a broader understanding of computing concepts are encouraged to also take the lecture.

Course Learning Outcomes

Students who successfully complete this course will be able to:

1. Demonstrate an understanding of how computers work, including a overview of hardware, logic gates, binary, and operating systems.
2. Demonstrate an understanding of cloud computing, cyber security principles, basic networking.
3. Demonstrate an understanding of Data Analysis using spreadsheets, databases and data visualization.
4. Demonstrate an understanding of the basics of programming.
5. Use computational thinking to ask questions and find different ways to solve problems.

CSE 1300: Introduction to Computing Principles

3 Credit Hours

Description

This course is an introductory computing principles course. Instruction centers on an overview of the history, scope, and impact of computing as well as critical, algorithmic and computational thinking on problem decomposition and fundamental programming concepts.

CSE 1321: Programming and Problem Solving I

3 Credit Hours

Concurrent: CSE 1321L

Description

This course provides an introduction to computing with a focus on programming. Instruction centers on an overview of programming, problem-solving, and algorithm development. Particular topics include object-oriented design/programming, primitive data types, arithmetic and logical operators, selection and repetition structures, interactive user input, exception handling, using and designing basic classes, single-dimensional data structures with searching and sorting, and arrays. Programming assignments focus on techniques of good programming style including proper documentation. The student is taught to efficiently design, code, and debug problem solutions and the relationship between correct code and security.

Course Learning Outcomes

Students who successfully complete this course will be able to:

1. Apply computational methods to analyze data and solve real-world problems.
2. Demonstrate the appropriate use of data types and logical, relational, and arithmetic expressions in programs.
3. Construct algorithmic solutions that incorporate selection and repetition.

4. Implement collections appropriately to store and analyze data.
5. Define functions and classes in program solutions.

- ## CSE 1321L: Programming and Problem Solving I Laboratory

1 Credit Hours

Concurrent: CSE 1321
Description
Lab activities with programming to accompany CSE 1321.

Course Learning Outcomes
Students who successfully complete this course will be able to:

1. Apply computational methods to analyze data and solve real-world problems.
2. Demonstrate knowledge of the syntax of a specific programming language.
3. Employ IDEs and compilers to learn how to compile, debug, run and test programs.
4. Demonstrate the appropriate use of data types and logical, relational, and arithmetic expressions in programs.
5. Construct algorithmic solutions that incorporate selection and repetition.
6. Implement collections appropriately to store and analyze data.
7. Define functions and classes in program solutions.

- ## CSE 1322: Programming and Problem Solving II

3 Credit Hours

Prerequisite: CSE 1321 and CSE 1321L
Concurrent: CSE 1322L and (MATH 1113 or MATH 1190 or MATH 1179 or MATH 2202)
Description
The second course in computing provides coverage of more advanced topics of object-oriented programming. This includes the use of static variables and classes, non-linear data structures, inheritance and polymorphism, file input/output, exception handling, recursion, and parameterized types. Elementary data structures (linked lists, stacks, and queues) are introduced to solve application problems. Graphical user interfaces, parallel programming, database programming, and event-driven programming are also introduced. Students will use good programming style including proper documentation.

- ## CSE 1322L: Programming and Problem Solving II Laboratory

1 Credit Hours

Prerequisite: CSE 1321 and CSE 1321L
Concurrent: CSE 1322
Description
This course is the required and supervised lab course to accompany CSE 1322.

- ## CSE 2300: Discrete Structures for Computing

3 Credit Hours

Prerequisite: ((CSE 1321 and CSE 1321L with a grade of "B" or higher) and (MATH 1113 or MATH 1190 or MATH 1179)
Description
Coverage of discrete structures is crucial to any program in computing. This course covers propositional and predicate logic, proofs, set theory, relations and functions, algorithms and complexity theory, matrices, graphs and trees, and combinatorics. Throughout, the emphasis will be on applications of these concepts in computing.

- ## CSE 3153: Database Systems

3 Credit Hours

Prerequisite: (CSE 1322 and CSE 1322L) or IT 1113 or (IT 1114 and IT 1114L)
Description
The topics in this course span from a review of the traditional file processing systems to database management systems. Topics include files systems and file processing logic, planning, and major phases of database development: analysis, design and implementation. Labs use an SQL based database product such as Oracle.

- ## CSE 3801: Professional Practices and Ethics

2 Credit Hours

Prerequisite: (CSE 1322 and CSE 1322L) or IT 3123 (may take concurrently)
Description
This course covers the historical, social and economic consideration of the discipline. It includes studies of professional conduct, risks, and liabilities, and intellectual property relative to the software engineering and computing professions. Software engineering/computing case studies will be used.

- ## CSE 4973: Technology Innovation and Entrepreneurship

3 Credit Hours

Prerequisite: ENTR 3001

Description

In this course, students will learn how to apply entrepreneurial thinking to technology innovation and commercialization. Topics of this course include practicing customer-centric innovation in IT and computing, matching customer needs with innovation seeds, evaluating readiness and market fit, building an IT startup team, creating a winning business model, and developing an investor pitch for the innovative solution.

- ## CSE 4983: CSE Computing Internship

3 Credit Hours

Prerequisite: Senior standing

Description

This course helps students gain practical experience through real-world projects and professional work. Students demonstrate an ability to apply computing principles and technologies relevant to their major in a specific real-world project jointly supervised by an industry mentor and a faculty advisor. Students work in a project team in an enterprise environment demonstrating ethical behavior as a computing professional, an understanding of social, professional and ethical issues related to computing, and an ability to integrate the knowledge acquired in preceding courses. Communication skills and leadership are also evaluated as well as professional computing skills and knowledge. 150+ hours per semester required at an internship site. The course can not be repeated for credit.

Contract All Courses |